



Student No:	

Mathematics Extension 2

Year 12, Assessment task 3, Term 2 2024

General Instruction:

- Reading time – 10 minutes
- Working time – 180 minutes
- Write using black/blue pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations

Total marks:

Section I – 10 marks (pages 1–3)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 4–9)

- Attempt Questions 11–16
- Allow about 2 hour and 35 minutes for this section

Class Teacher (please tick one name)

- ☐ Mr Berry
- ☐ Ms Lee
- ☐ Mr Umakanthan
- ☐ Dr Vranešević

Question No.	1-10	11	12	13	14	15	16	Total
Marks	/10	/15	/15	/15	/15	/15	/15	/100

Section I**10 marks****Attempt Questions 1–10****Allow about 15 minutes for this section**

Use the multiple-choice answer sheet for Questions 1–10.

1. Which of the following is **NOT** equivalent to ‘If n is even, then n^2 is a multiple of 4’?

- A) n is even $\therefore n^2$ is a multiple of 4;
- B) n^2 is a multiple of 4 if n is even;
- C) n^2 is a multiple of 4 $\therefore n$ is even;
- D) n is even implies that n^2 is a multiple of 4.

2. Let $I_n = \int x^n e^{ax} dx$. Which of the following is the correct expression for I_n ?

- A) $I_n = \frac{x^n e^{ax}}{a} - nI_{n-1}$;
- B) $I_n = \frac{x^n e^{ax}}{a} - \frac{n}{a}I_{n-1}$;
- C) $I_n = \frac{x^n e^{ax}}{a} + nI_{n-1}$;
- D) $I_n = \frac{x^n e^{ax}}{a} + \frac{n}{a}I_{n-1}$.

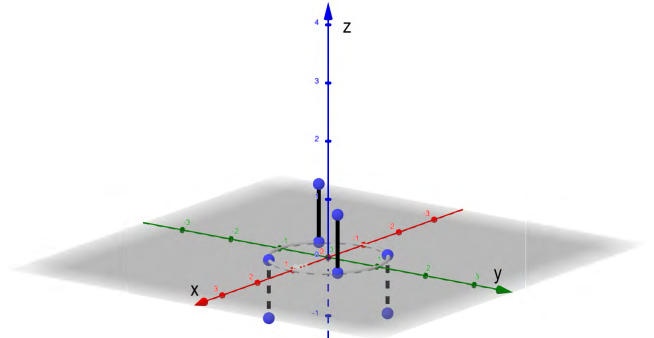
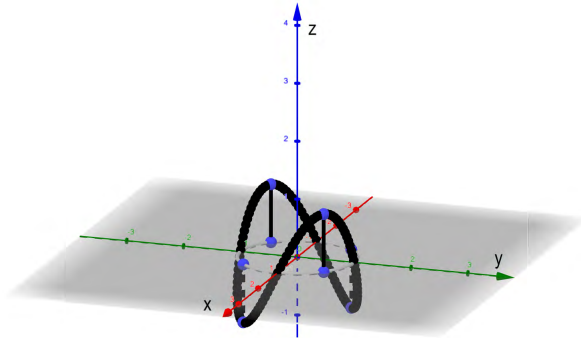
3. The acceleration of a particle moving in a straight line with velocity v is given by $\ddot{x} = v^2$. Initially $v = 1$. What is v as a function of t ?

- A) $v = 1 - t$;
- B) $v = \ln|1 - t|$;
- C) $v = \frac{t}{1 - t}$;
- D) $v = \frac{1}{1 - t}$.

4. The negation of the statement $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}$ such that $2x + 3y = 12$ is:

- A) $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}$ such that $2x + 3y \neq 12$;
- B) $\exists x \in \mathbb{R}$ such that $\forall y \in \mathbb{R} 2x + 3y \neq 12$;
- C) $\forall x \in \mathbb{R}, \forall y \in \mathbb{R} 2x + 3y \neq 12$;
- D) $\exists x \in \mathbb{R}, \exists y \in \mathbb{R}$ such that $2x + 3y \neq 12$.

5. The sketch of the curve in the shape of saddle is drawn on the left. The curve has two peaks at $t = \frac{\pi}{4}, \frac{5\pi}{4}$ and two troughs at $t = \frac{3\pi}{4}, \frac{7\pi}{4}$, shown on the right. Which of following parametric equations are represented by the given curve:



- A) $x = \sin t, y = \cos 2t, z = \sin t$;
 B) $x = \sin t, y = \cos t, z = \sin 2t$;
 C) $x = \sin 2t, y = \cos t, z = \sin t$;
 D) $x = \sin 2t, y = \cos t, z = \sin 2t$.
6. Which of the following complex numbers equals $(\sqrt{3} + i)^4$
- A) $-2 + \frac{2}{\sqrt{3}}i$;
 B) $-8 + \frac{8}{\sqrt{3}}i$;
 C) $-2 + 2\sqrt{3}i$;
 D) $-8 + 8\sqrt{3}i$.
7. The line l_1 has vector equation $\mathbf{r}_1 = \mathbf{i} + \lambda(\mathbf{j} - \mathbf{k})$ and the line l_2 has vector equation $\mathbf{r}_2 = (3\mathbf{i} + 2\mathbf{j} - \mathbf{k}) + \mu(2\mathbf{i} + \mathbf{k})$. Which of the following statements is correct?
- A) l_1 and l_2 are parallel;
 B) l_1 and l_2 are perpendicular;
 C) l_1 and l_2 intersect at a point;
 D) l_1 and l_2 are skew.
8. Which of the following is an expression for $\int \frac{4x^2 - 5x - 1}{(x - 3)(x^2 + 1)} dx$
- A) $\ln[(x - 3)(x^2 + 1)] + C$;
 B) $\ln[(x - 3)^2(x^2 + 1)] + C$;
 C) $\ln[(x - 3)(x^2 + 1)] + \tan^{-1} x + C$;
 D) $\ln[(x - 3)^2(x^2 + 1)] + \tan^{-1} x + C$.

9. What is the solution to the equation $z^2 = i\bar{z}$

- A) $(0,0)$ and $(0,1)$;
- B) $(0,0)$ and $(0,-1)$;
- C) $(0,0)$, $(0,-1)$, $(-\frac{\sqrt{3}}{2}, \frac{1}{2})$ and $(\frac{\sqrt{3}}{2}, \frac{1}{2})$;
- D) $(0,0)$, $(0,1)$, $(-\frac{\sqrt{3}}{2}, \frac{1}{2})$ and $(\frac{\sqrt{3}}{2}, \frac{1}{2})$;

10. A particle of unit mass m is projected vertically upwards with an initial velocity of $u \text{ ms}^{-1}$ in a medium in which the resistance to the motion is proportional to the square of the velocity $v \text{ ms}^{-1}$ of the particle or mkv^2 . Let x be the displacement in metres of the particle above the point of projection, O, so that the equation of motion is $\ddot{x} = -(g + kv^2)$ where $g \text{ ms}^{-2}$ is the acceleration due to gravity and k is constant proportional to g , given as $k = 0.01g$.

Which of the following gives the correct expression for the time taken?

- A) $t = \frac{100}{g} \left(\tan^{-1} \frac{u}{10} - \tan^{-1} \frac{v}{10} \right)$;
- B) $t = \frac{10}{g} \left(\tan^{-1} u - \tan^{-1} v \right)$;
- C) $t = \frac{10}{g} \left(\tan^{-1} \frac{u}{10} - \tan^{-1} \frac{v}{10} \right)$;
- D) $t = \frac{100}{g} \left(\tan^{-1} u - \tan^{-1} v \right)$.

End of Section I

Section II**90 marks****Attempt Questions 11–16****Allow about 2 hour and 35 minutes for this section**

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 11. (15 marks) Use the Question 11 Writing Booklet

(a) State whether each statement is true or false, justifying your answer.

(i) $\forall n \in \mathbb{N}, n^2 > n$ 1

(ii) $\exists x \in \mathbb{R}$ such as $x^2 + 1 = 0$ 1

(iii) $\forall x \in \mathbb{R}, \exists y \in \mathbb{N}$ such as $x + y = 10$ 1

(b) It is given that $f(z) = z^4 + 2\sqrt{2}z^3 + z^2 + 8\sqrt{2}z - 12$. One of the roots of the equation $f(z) = 0$ is given by $2i$. By factorising $f(z)$ as a product of two quadratic factors, obtain the other roots of the equation.

3

(c) Find the integral:

$$\int \frac{e^{\arctan x} + x \ln(1+x^2) + 1}{1+x^2} dx$$
 2

(d) The displacement of a particle moving in a straight line, s m, at time x seconds is given by $s = \sin 4x + 2 \sin 2x + 2$.

(i) Show that the velocity of the particle at time x seconds is given by

$$v = \frac{8}{(1+t^2)^2} (1-3t^2) \text{ ms}^{-1}, \text{ where } t = \tan x$$
 2

(ii) Hence find the value of x where $0 \leq x \leq \pi$ for which the displacement is maximised. 2

(e) Find the solutions of the integral equation with respect to a , for $2 \leq a \leq 3$, of

$$\int_0^a \cos(x+a^2) dx = \sin a$$
 3

Question 12. (15 marks) Use the Question 12 Writing Booklet

- (a) A complex number z_1 is given by $z_1 = a + (a - 3)i$, where a is a positive real constant. It is given that $\arg z_1 = \theta$, where $-\frac{\pi}{2} < \theta < 0$.

- (i) Find, leaving your answer in terms of θ ,

(α) $\arg(-2z_1)$, 1

(β) $\arg(z_1 - 2a)$. 2

- (ii) Another complex number z_2 is given by $z_2 = 1 + 3i$. Without using a calculator, find the range of values of a such that

$$\frac{|z_1 z_2|^2}{\operatorname{Im}(z_1 z_2)} \leq 10$$
4

- (b) Prove that a four-digit number is divisible by 11 if and only if the difference between the sum of its even digits and the sum of its odd digits is a multiple of 11 (including zero).

Hint: Divide 1001 by 11.

3

- (c) (i) Differentiate $\cot^3 x$ 1

- (ii) Differentiate $\cot^5 x$ 1

- (iii) Using derivatives from (i) and (ii) find the integral of

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^6 x \, dx$$
3

Question 13. (15 marks) Use the Question 13 Writing Booklet

- (a) (i) if k is an integer where $k \geq 3$ and $(k-1)(k+1) < k^2$, show that

$$\frac{1}{(k-1)k(k+1)} > \frac{1}{k^3} \quad 1$$

- (ii) Given that $S_n = \frac{1}{3^3} + \frac{1}{4^3} + \frac{1}{5^3} + \dots + \frac{1}{n^3} = \sum_3^n \frac{1}{k^3}$, use partial fractions from previous part or otherwise to prove that $S_n < \frac{1}{12}$. 3

- (b) Let p and q be positive real numbers with $\frac{1}{p} + \frac{1}{q} = 1$. Prove that

(i)

$$\frac{1}{3} \leq \frac{1}{p(p+1)} + \frac{1}{q(q+1)} \leq \frac{1}{2} \quad 2$$

(ii)

$$\frac{1}{p(p-1)} + \frac{1}{q(q-1)} \geq 1 \quad 2$$

- (c) An object is moving on a horizontal plane and at position x metres from the origin it has acceleration ($\ddot{x} \text{ m s}^{-2}$) given by $\ddot{x} = 0.01 \left(x + \frac{64}{x^3} \right)$. Initially the object is 4 metres to the left of the origin and moving to the left at a speed of $\frac{\sqrt{3}}{5} \text{ m s}^{-1}$.

- (i) If the velocity of the object is $v \text{ m s}^{-1}$, show that

$$v^2 = 0.01 \left(\frac{x^4 - 64}{x^2} \right). \quad 2$$

- (ii) Explain why the velocity of the object at a position x metres from the origin is given by

$$v = \frac{\sqrt{x^4 - 64}}{10x}. \quad 1$$

- (iii) Find, correct to the nearest second, the time to reach a position 50 metres to the left of the origin. 4

Question 14. (15 marks) Use the Question 14 Writing Booklet

- (a) Let $a_1, a_2, \dots, a_n$ be positive real numbers such that $a_1 a_2 \dots a_n = 1$. Prove that

$$(a_1^2 + a_1)(a_2^2 + a_2) \dots (a_n^2 + a_n) \geq 2^n.$$

3

- (b) (i) A parachutist of mass 90 kg jumps out of an aeroplane at a height of 100 m. The acceleration due to gravity is 10 m/s^2 .

Derive the vertical velocity functions $v \text{ m/s}^2$ in terms of displacement x for the parachutist, if the initial velocity is $u \text{ m/s}^2$ and there is:

(α) **no** resistance force during free fall; **2**

(β) a resistance force of $0.27v^2$ Newtons when parachute is open. **3**

- (ii) A parachutist jumps from rest, and opens his parachute 60 m from the ground.

(α) Find the vertical velocity when parachute is open. **1**

(β) Find the vertical velocity (1 decimal place) on landing. **2**

(γ) What percentage (to the nearest whole percent) is the landing velocity compared to the terminal velocity? **2**

- (c) Let a, b, c be positive integers. Prove that it is impossible for all three numbers $a^2 + b + c$, $b^2 + a + c$, $c^2 + a + b$ to be perfect squares.

2

Question 15. (15 marks) Use the Question 15 Writing Booklet

(a) Consider the following vector equation $\mathbf{r} = (2 + \lambda)\mathbf{i} + (3 - \lambda)\mathbf{j} + (4 - 2\lambda)\mathbf{k}$.

(i) Find the coordinates of the points where the line $\mathbf{r}$ cuts the coordinate planes, where point A is on xy plane, point B is on xz plane, and point C is on yz plane. 3

(ii) Find the vector projection of $\overrightarrow{OC}$ onto $\overrightarrow{AB}$ 1

(iii) Find the area of $\triangle OAC$ correct to 3 significant figures. 2

(b) Let $I_n = \int_0^{\frac{\pi}{2}} (\sin x)^n dx$, where n is an integer $n \geq 0$.

(i) Using integration by parts, show that for $n \geq 2$, 3

$$I_n = \left(\frac{n-1}{n}\right) I_{n-2}$$

(ii) Deduce that 3

$$I_{2n} = \frac{2n-1}{2n} \cdot \frac{2n-3}{2n-2} \cdots \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} \quad \text{and} \quad I_{2n+1} = \frac{2n}{2n+1} \cdot \frac{2n-2}{2n-1} \cdots \frac{4}{5} \cdot \frac{2}{3} \cdot 1$$

(iii) Explain why 1

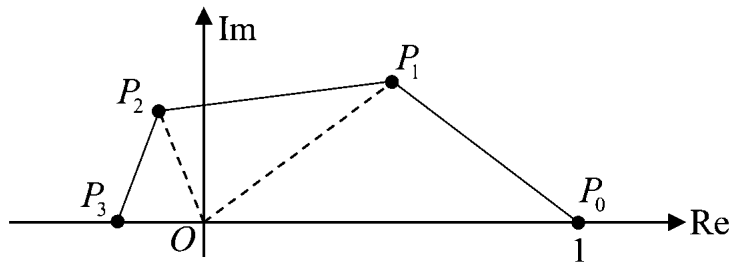
$$I_k > I_{k+1}$$

(iv) Hence, using the fact that $I_{2n-1} > I_{2n}$ and $I_{2n} > I_{2n+1}$, show that 2

$$\frac{\pi}{2} \left(\frac{2n}{2n+1} \right) < \frac{2^2 \cdot 4^2 \cdots (2n)^2}{1 \cdot 3^2 \cdot 5^2 \cdots (2n-1)^2 \cdot (2n+1)} < \frac{\pi}{2}$$

Question 16. (15 marks) Use the Question 16 Writing Booklet

- (a) A complex number z is given by $z = re^{i\frac{\pi}{n}}$, where $0 < r < 1$ and n is a positive integer with $n \geq 3$. The numbers $1, z, z^2, \dots, z^n$ can be represented by the points $P_0, P_1, P_2, \dots, P_n$ respectively in an Argand diagram. The $(n+1)$ -sided polygon formed by using $P_0, P_1, P_2, \dots, P_n$ is called the $(n+1)$ -polygon generated by z . An example of a $(3+1)$ -polygon generated by z is shown in the following Argand diagram (not drawn to scale).



Let $z = \frac{1}{4}(1 + \sqrt{3}i)$ for parts (i) to (iii).

- (i) Express z in the form $re^{i\theta}$ where $r > 0$ and $0 < \theta \leq \pi$. 2
- (ii) Hence write down z^2 and z^3 in similar form. 2
- (iii) Find the exact area of triangle OP_0P_1 , where O is the origin. Hence find the exact area of the $(3+1)$ -polygon generated by z . 4

Let $z = \frac{1}{2}re^{i\frac{\pi}{n}}$ for parts (iv)

- (iv) Find the area of an $(n+1)$ -polygon generated by z in terms of n , leaving your answer in the form $a(1 - b^n) \sin \frac{\pi}{n}$, where a and b are real numbers to be determined. 3
- (b) (i) Consider triangle ABC , where $\overrightarrow{AB} = \mathbf{b}$ and $\overrightarrow{AC} = \mathbf{c}$. Find a vector (in terms of $\mathbf{b}$ and $\mathbf{c}$) which bisects the angle $\angle CAB$. 1
- (ii) On the same triangle ABC , D and E are points on the sides AC and AB of the triangle, respectively. Also, DE is not parallel to CB . Suppose F and G are points of BC and ED , respectively, such that $BF : FC = EG : GD = BE : CD$. Show that GF is parallel to the angle bisector of $\angle CAB$. 3

End of Examination

Q1 A) n is even $\Rightarrow n^2$ is multiple of 4

B) n^2 is a multiple of 4 if n is even

C) n^2 is a multiple of 4 $\Rightarrow n$ is even

D) n is even implies n^2 is a multiple of 4

$\therefore$ (C)

Q2 $I_n = \int x^n e^{ax} dx$

$$u = x^n \quad v' = e^{ax}$$

$$u' = nx^{n-1} \quad v = \frac{1}{a} e^{ax}$$

(by parts)

$$= \frac{x^n e^{ax}}{a} - \int nx^{n-1} \cdot \frac{1}{a} e^{ax} dx$$

$$= \frac{x^n e^{ax}}{a} - \frac{n}{a} \int x^{n-1} e^{ax} dx$$

$$\therefore I_n = \frac{x^n e^{ax}}{a} - \frac{n}{a} I_{n-1}$$

(B)

Q3 $\ddot{x} = v^2$, $t=0, v=1$

$$v = f(t) = ?$$

$$\ddot{x} = \frac{dv}{dt} = v^2$$

(diff. equation of motion)

$$\frac{dt}{t} = \frac{dv}{v^2} \quad \text{Q3 continues}$$

$$\int_0^t \frac{dt}{t} = \int_1^{v^{-2}} dv$$

$$t = (-1) v^{-1} \Big|_1^v$$

$$= (-1) \left(\frac{1}{v} - \frac{1}{1} \right)$$

$$= (-1) \left(\frac{1}{v} - 1 \right)$$

$$t = \frac{v-1}{v}$$

$$tv = v - 1$$

$$v(t-1) = -1$$

$$v = \frac{1}{1-t}$$

⑦

Q4. $\neg (\forall x \in \mathbb{R}, \exists y \in \mathbb{R} \text{ such that } 2x+3y=12)$

$\exists x \in \mathbb{R} \text{ such that } \forall y \in \mathbb{R} \ 2x+3y \neq 12$

$\therefore$ ⑧ (quantifiers)

Q6. $(\sqrt{3}+i)^4 = \left[2 \left(\frac{\sqrt{3}}{2} + i \frac{1}{2} \right) \right]^4$
 $= 2^4 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right)^4$
 $= 2^4 \left(\cos \frac{4\pi}{6} + i \sin \frac{4\pi}{6} \right)$ (CN-powers)
 $= 16 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right)$ De-M
 $= 16 \left(-\frac{1}{2} + i \frac{\sqrt{3}}{2} \right)$
 $= -8 + 8\sqrt{3}i$

①

Q7 $\vec{r}_1 = \hat{i} + \lambda (\hat{j} - \hat{k}) = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$
 $\vec{r}_2 = \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}$ (vector equation of straight line)
 $\begin{pmatrix} 1 \\ \lambda \\ -\lambda \end{pmatrix} = \begin{pmatrix} 3+2\mu \\ 2 \\ -1+\mu \end{pmatrix} \Rightarrow \lambda = 2$
 $-2 = -1 + \mu \Rightarrow \mu = -1$
 $1 = 3 + 2(-1) = 1$
 $\therefore \vec{r}_1 \text{ \& } \vec{r}_2 \text{ has a point of intersection}$

②

Q5 for $t = \frac{\pi}{4}$

A) $x = \frac{\sqrt{2}}{2}, y = 1, z = \frac{\sqrt{2}}{2} \times$

B) $x = \frac{\sqrt{2}}{2}, y = \frac{\sqrt{2}}{2}, z = 1$
 $t = \frac{5\pi}{4} : x = -\frac{\sqrt{2}}{2}, y = -\frac{\sqrt{2}}{2}, z = 1$ } for peaks
 $t = \frac{3\pi}{4} : x = -\frac{\sqrt{2}}{2}, y = \frac{\sqrt{2}}{2}, z = -1$
 $t = \frac{7\pi}{4} : x = \frac{\sqrt{2}}{2}, y = -\frac{\sqrt{2}}{2}, z = -1$ } for troughs

$\therefore$ **B**

Q8 $\int \frac{4x^2 - 5x - 1}{(x-3)(x^2+1)} dx$ (partial fraction)

$$\frac{4x^2 - 5x - 1}{(x-3)(x^2+1)} = \frac{a}{x-3} + \frac{bx+1}{x^2+1}$$

$$4x^2 - 5x - 1 = a(x^2+1) + (bx+1)(x-3)$$

$$= (a+b)x^2 + (1-3b)x + a-3$$

$$a+b=4 \Rightarrow$$

$$-5 = 1-3b \Rightarrow -6 = -3b \Rightarrow b = 2$$

$$-1 = a-3 \Rightarrow a = 2$$

$$\therefore \int \left(\frac{2}{x-3} + \frac{2x+1}{x^2+1} \right) dx =$$

$$= \int \frac{2}{x-3} dx + \int \frac{2x dx}{x^2+1} + \int \frac{dx}{1+x^2}$$

$$= 2 \ln|x-3| + \ln|x^2+1| + \tan^{-1} x + C$$

$$= \ln[(x^2+1)(x-3)^2] + \tan^{-1} x + C$$

$$\therefore \textcircled{D}$$

Q9. $z^2 = i\bar{z}$ $z = x+iy$ $\bar{z} = x-iy$

$$z^2 = i\bar{z}$$

$$(x+iy)^2 = i(x-iy)$$

(CN arithmetics equations)

$$x^2 - y^2 + 2xyi = +y + xi$$

$$x^2 - y^2 = y \quad \uparrow$$

$$2xy = x \Rightarrow x(2y-1) = 0 \Rightarrow \underline{x=0 \text{ or } y=\frac{1}{2}}$$

Sub into (i)

$$\text{for } x=0 : -y^2 = y \\ y(1+y)=0 \Rightarrow y=0 \text{ or } y=-1$$

$$\text{for } y=\frac{1}{2} : x^2 - \frac{1}{4} = \frac{1}{2} \\ x^2 = \frac{3}{4} \Rightarrow x = \pm \frac{\sqrt{3}}{2}$$

$$\therefore (0,0), (0,-1), \left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right), \left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$$

$\therefore \textcircled{C}$

$$\text{Q10 } m=1, v_0=u, R=ukv^2, g, k=0.01g$$

$$m\ddot{x} = -mg - mkv^2$$

$$\ddot{x} = -(g + 0.01gv^2)$$

$$\frac{dv}{dt} = -g(1 + 0.01v^2)$$

$$\frac{dt}{dv} = \frac{-1}{g} \cdot \frac{1}{(1 + 0.01v^2)}, \quad (0.1)^2 = 0.01$$

$$\int_0^t dt = -\frac{1}{g} \int \frac{dv}{1 + (0.1v)^2}$$

$$\begin{aligned}
 t &= -\frac{1}{g} \frac{1}{0.1} \int_u^v \frac{0.1 \, dv}{1 + (0.1v)^2} \\
 &= -\frac{10}{g} \left[\tan^{-1} 0.1v \right]_u^v = -\frac{10}{g} \left[\tan^{-1} \frac{v}{10} \right]_u^v \\
 &= -\frac{10}{g} \left[\tan^{-1} \frac{v}{10} - \tan^{-1} \frac{u}{10} \right] \\
 &= \frac{10}{g} \left(\tan^{-1} \frac{u}{10} - \tan^{-1} \frac{v}{10} \right)
 \end{aligned}$$

$\therefore (C)$

Q₁ C

Q₂ B

Q₃ D

Q₄ B

Q₅ B

Q₆ D

Q₇ C

Q₈ D

Q₉ C

Q₁₀ C

Q 11 a)

i) $\forall n \in \mathbb{N}, n^2 > n$ False as for $n=1$, $1 > 1$ not true ①

ii) $\exists x \in \mathbb{R}$, such as $x^2 + 1 = 0$ False as ①
 $x^2 = -1 \Rightarrow x = \sqrt{-1} = \pm i \notin \mathbb{R}$

iii) $\forall x \in \mathbb{R}, \exists y \in \mathbb{N}$ such as $x+y=10$ False ①
as for $x = \frac{1}{2}$, $y = 9\frac{1}{2} \notin \mathbb{N}$

b) $f(z) = z^4 + 2\sqrt{2}z^3 + z^2 + 8\sqrt{2}z - 12$

$f(z) = 0$ for $z = 2i$ find other roots

since all the coeff. of $f(z)$ are real: then
 $f(2i) = 0$, then $f(-2i) = 0 \therefore -2i$ is another root ①

Thus product of quadratic factors =
 $[z - (2i)][z - (-2i)]$
 $= z^2 + 4$

$\therefore f(z) = (z^2 + 4)(z^2 + Az - 3)$
 $= z^4 + 4z^2 + Az^3 + 4Az - 3z^2 - 12$
 $= z^4 + Az^3 + z^2 + 4Az - 12$
 $\therefore \begin{cases} A = 2\sqrt{2} \\ 4A = 8\sqrt{2} \end{cases} \Rightarrow A = 2\sqrt{2}$ ①

$$\Rightarrow A = 2\sqrt{2}$$

$$\therefore f(z) = (z^2 + 4)(z^2 + 2\sqrt{2}z - 3)$$

$$z^2 + 2\sqrt{2}z - 3 = 0$$

$$z_{1,2} = \frac{-2\sqrt{2} \pm \sqrt{8 - 4 \times 1 \times (-3)}}{2}$$

$$= \frac{-2\sqrt{2} \pm \sqrt{8+12}}{2}$$

$$= \frac{-2\sqrt{2} \pm 2\sqrt{5}}{2}$$

$$= -\sqrt{2} \pm \sqrt{5}$$

①

$\therefore$ roots are $2i, -2i, -\sqrt{2} + \sqrt{5}, -\sqrt{2} - \sqrt{5}$

$$c) \int \frac{e^{\arctan x} + x \ln(1+x^2) + 1}{1+x^2} dx =$$

$$= \int \frac{e^{\arctan x}}{1+x^2} dx + \int \frac{x \ln(1+x^2)}{1+x^2} dx + \int \frac{1}{1+x^2} dx$$

Sub. $\arctan x = t$
 $\frac{dx}{1+x^2} = dt$

$\therefore \ln(1+x^2) = u$
 $\frac{2x dx}{1+x^2} = du$ (1)

$$= \int e^t dt + \frac{1}{2} \int u du + \arctan x + C$$

$$= e^t + \frac{1}{2} \frac{u^2}{2} + \arctan x + C$$

$$= e^{\arctan x} + \frac{1}{4} [\ln(1+x^2)]^2 + \arctan x + C$$

d) $s(x) = \sin 4x + 2 \sin 2x + 2$
 x -time [sec]

i) $v = \frac{8}{(1+t^2)^2} (1-3t^2) \frac{m}{s}, t = \tan x$

$$\frac{ds}{dx} = 4 \cos 4x + 4 \cos 2x$$

$$t = \tan x$$

$$\cos 2x = \frac{1-t^2}{1+t^2} \quad \sin 2x = \frac{2t}{1+t^2} \quad \textcircled{1}$$

$$\cos 2x \cdot 2x = \cos^2 2x - \sin^2 2x$$

$$\therefore \frac{ds}{dx} = v = 4x \left(\left(\frac{1-t^2}{1+t^2} \right)^2 - \left(\frac{2t}{1+t^2} \right)^2 \right)$$

$$+ 4x \frac{1-t^2}{1+t^2}$$

$$= 4 \frac{1-2t^2+t^4-4t^2}{(1+t^2)^2} + \frac{4-4t^2}{1+t^2}$$

$$= \frac{4-24t^2+4t^4+4+t^2-t^2-4t^2}{(1+t^2)^2}$$

$$= \frac{8 \cdot 24t^2}{(1+t^2)^2} = \frac{8(1-3t^2)}{(1+t^2)^2} \cdot \frac{4}{5} \quad \square \quad \textcircled{1}$$

ii) $0 < \alpha < \pi$ $S_{\max} = ?$

for $S_{\max}$, $\dot{y} = 0$

$$\therefore 1-3t^2 = 0 \Rightarrow t^2 = \frac{1}{3} \Rightarrow t = \pm \sqrt{\frac{1}{3}} = \pm \frac{\sqrt{3}}{3}$$

$$\pm \frac{\sqrt{3}}{3} = \tan \alpha \Rightarrow \alpha = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\frac{d^2s}{dx^2} = -16 \sin 4\alpha - 8 \sin 2\alpha \quad \textcircled{1}$$



$$\frac{d^2s}{dx^2} \left(\alpha = \frac{\pi}{6} \right) = -16 \frac{\sqrt{3}}{2} - 8 \frac{\sqrt{3}}{2} = -24 \frac{\sqrt{3}}{2} = -12\sqrt{3} < 0 \quad \checkmark$$

$$\frac{d^2s}{dx^2} \left(\alpha = \frac{5\pi}{6} \right) = -16 \left(-\frac{\sqrt{3}}{2} \right) - 8 \left(-\frac{\sqrt{3}}{2} \right) = 12\sqrt{3} > 0 \quad \checkmark$$

$\therefore$ max of $s(x)$ is for $x = \frac{\pi}{6} \quad \textcircled{1}$

$$e) \quad 2 \leq a \leq 3$$

$$\int_0^a \cos(x+a^2) dx = \sin a$$

$$\sin(x+a^2) \Big|_0^a = \sin a$$

$$\sin(a+a^2) - \sin a^2 = \sin a \quad (1)$$

using Product to sum trig. id.

$$2 \cos A \sin B = \sin(A+B) - \sin(A-B)$$

$$A+B = a+a^2$$

$$A-B = a^2$$

$$\Rightarrow B = A - a^2$$

$$= \frac{2a^2 + a - 2a}{2} = \frac{a}{2}$$

$$2A = 2a^2 + a$$

$$A = \frac{2a^2 + a}{2}$$

$\therefore$ (1) becomes

$$2 \cos \frac{2a^2+a}{2} \sin \frac{a}{2} = \sin a$$

then using double angle on right

$$2 \cos \frac{2a^2+a}{2} \sin \frac{a}{2} = 2 \sin \frac{a}{2} \cos \frac{a}{2}$$

$$2 \sin \frac{a}{2} \left[\cos \frac{2a^2+a}{2} - \cos \frac{a}{2} \right] = 0 \quad (1)$$

$$2 \sin \frac{a}{2}$$

$$2 \sin A \sin B = \cos(A-B) - \cos(A+B)$$

$$\left. \begin{array}{l} A-B = \frac{2a^2+a}{2} \\ A+B = \frac{a}{2} \end{array} \right\} B = \frac{a^2+a}{2} - \frac{2a^2+a}{2} = \frac{-a^2}{2}$$

$$2A = \frac{2a^2+2a}{2} = a^2+a$$

$$A = \frac{a^2+a}{2}$$

$$\therefore 2 \sin \frac{a}{2} \times 2 \sin \frac{a^2+a}{2} \sin \left(\frac{-a^2}{2} \right) = 0$$

$$-4 \sin \frac{a}{2} \sin \frac{a^2+a}{2} \sin \frac{a^2}{2} = 0$$

$$\sin \frac{a}{2} \sin \frac{a^2+a}{2} \sin \frac{a^2}{2} = 0$$

$$\therefore \frac{a}{2} = k\pi \text{ or } \frac{a^2+a}{2} = n\pi \text{ or } \frac{a^2}{2} = m\pi$$

$$a = 2k\pi \text{ or } a = \pm \sqrt{2m\pi} \text{ or } a = \frac{-1 \pm \sqrt{1+8n\pi}}{2}$$

$$a^2 + a - 2n\pi = 0$$

$$a = \frac{-1 \pm \sqrt{1 - 4 \times 1 \times (-2n\pi)}}{2}$$

$$a = \frac{-1 \pm \sqrt{1+8n\pi}}{2}$$

where $k, m, n \in \mathbb{Z}$

as $2 \leq a \leq 3$

all values
should be
positive

$$\text{now: } 2 \leq a \leq 3$$

$$\therefore 2 \leq 2k\pi \leq 3$$

$$\frac{1}{\pi} \leq k \leq \frac{3}{2\pi}$$

$$0.318 \leq k \leq 0.477$$

$$2 \leq \frac{-1 + \sqrt{1+8n\pi}}{2} \leq 3$$

$$5 \leq \sqrt{1+8n\pi} \leq 7$$

$$24 \leq 8n\pi \leq 48$$

$$\frac{24}{8\pi} \leq n \leq \frac{48}{8\pi}$$

$$2 \leq \sqrt{2m\pi} \leq 3$$

$$\frac{4}{2\pi} \leq m \leq \frac{9}{2\pi}$$

$$0.64 \leq m \leq 1.432$$

$$\therefore m = 1$$

$$\frac{3}{\pi} \leq n \leq \frac{6}{\pi}$$

$$0.95 \leq n \leq 1.91$$

$$\therefore \underline{\underline{n=1}}$$

Final solution for a is

$$a = \sqrt{2\pi} \text{ or } \frac{-1 + \sqrt{1+8\pi}}{2}$$

①

or in set notation

$$a \in \left\{ \sqrt{2\pi}, \frac{-1 + \sqrt{1+8\pi}}{2} \right\}$$

Q12

a) $z_1 = a + (a-3)i$, $a \in \mathbb{R}^+$

i) $\arg z_1 = \theta$, $-\frac{\pi}{2} < \theta < 0$

$\arg(-2z_1) = \arg(-2) + \arg(z_1)$
 $= \pi + \theta$ ①

b) $\arg(z_1 - 2a) = ?$

$z_1 - 2a = a + (a-3)i - 2a = -a + (a-3)i$

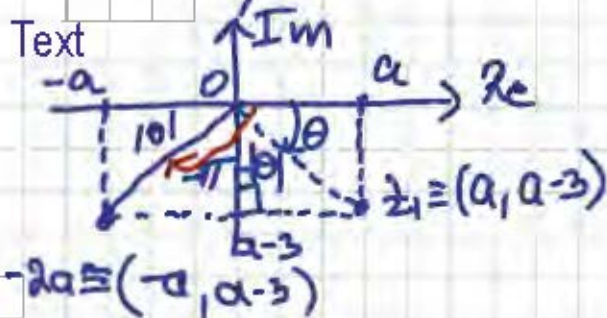
$\arg(z_1 - 2a) = \arg(-a + (a-3)i)$ ①

$= -\pi + |\theta|$

$= -\pi + (-\theta)$

$= -\pi + \theta$ ①

Text



ii) $z_2 = 1+3i$, find a such that

$$\frac{|z_1 z_2|^2}{\text{Im}(z_1 z_2)} \leq 10$$

$z_1 z_2 = (a + (a-3)i)(1+3i)$

$= (a - 3a + 9) + (3a + a - 3)i$

$= (-2a + 9) + (4a - 3)i$

$$\therefore \operatorname{Im}(z_1 z_2) = 4a - 3 \quad (1)$$

$$|z_1 z_2|^2 = |z_1|^2 |z_2|^2 = (a^2 + (a-3)^2)(1^2 + 3^2) \\ = 10(2a^2 - 6a + 9) \quad (1)$$

$$\frac{|z_1 z_2|^2}{\operatorname{Im}(z_1 z_2)} \leq 10$$

$$\frac{10(2a^2 - 6a + 9)}{4a - 3} \leq 10$$

$$\frac{2a^2 - 6a + 9}{4a - 3} \leq 1$$

$$\frac{2a^2 - 6a + 9 - (4a - 3)}{4a - 3} \leq 0$$

$$\frac{2a^2 - 6a + 9 - 4a + 3}{4a - 3} \leq 0$$

$$\frac{2a^2 - 10a + 12}{4a - 3} \leq 0$$

$$\frac{2(a-2)(a-3)}{4a-3} \leq 0 \quad (1)$$



$$\therefore a < \frac{3}{4} \text{ or } 2 \leq a < 3 \quad (1)$$

Since $-\frac{\pi}{2} < \arg z_1 < 0 \therefore \operatorname{Im}(z_1) = a - 3 < 0$ $2 \leq a < 3$
 also it is given $a > 0 \therefore 0 < a < 3 \therefore 0 < a < \frac{3}{4}$ or

b) let the digit of the number, in order, be a, b, c, d

$$N = 1000a + 100b + 10c + d \quad (1)$$

If N divisible by 11, then $1000a + 100b + 10c + d = 11k$
 $k \in \mathbb{Z}$

$$1001 \div 11 = 91$$

$$\therefore 1001a - a + 99b + b + 11c - c + d = 11k$$

$$11 \times 91a - a + 11 \times 9b + b + 11c - c + d = 11k$$

$$11 \times 91a + 11 \times 9b + 11c - 11k = a - b + c - d \quad (1)$$

$$11(91a + 9b + c - k) = (a + b) - (b + d)$$

$\therefore$ If N divisible by 11, the difference of sum of even digits and the sum of odd digits is a multiple of 11. (1)

Q12c) $\cot^3 x$, $\cot^5 x$
 $\int_{\pi/4}^{\pi/2} \cot^6 x dx = ?$

$\sin^2 x + \cos^2 x = 1$
 $1 + \cot^2 x = \operatorname{cosec}^2 x$

Let $y = \cot^5 x$
 $y' = -5 \cot^4 x \operatorname{cosec}^2 x$
 $= -5 \cot^4 x (1 + \cot^2 x)$
 $= -5 \cot^4 x - 5 \cot^6 x$

①

$5 \cot^6 x = -5 \cot^4 x - y$
 $\cot^6 x = -\frac{1}{5} y' - \cot^4 x$
 $\therefore \int_{\pi/4}^{\pi/2} \cot^6 x dx = - \left[\frac{1}{5} \int_{\pi/4}^{\pi/2} y' + \int_{\pi/4}^{\pi/2} \cot^4 x dx \right]$
 $= -\frac{1}{5} y \Big|_{\pi/4}^{\pi/2} - \int_{\pi/4}^{\pi/2} \cot^4 x dx$
 $= -\frac{1}{5} \cot^5 x \Big|_{\pi/4}^{\pi/2} - \int_{\pi/4}^{\pi/2} \cot^4 x dx$

$$= -\frac{1}{5} (0-1) - I \quad \textcircled{1}$$

$$I = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^4 x \, dx$$

$$y = \cot^3 x$$

$$y' = -3\cot^2 x \operatorname{cosec} x = 3\cot^2 x (1 + \cot^2 x) \quad \textcircled{1}$$

$$= -3\cot^2 x - 3\cot^4 x$$

$$\therefore \cot^4 x = -\frac{1}{3} y' - \cot^2 x$$

$$\therefore \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^4 x \, dx = -\frac{1}{3} y \Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}} - \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^2 x \, dx$$

$$= -\frac{1}{3} \cot^3 x \Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}} - \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (\operatorname{cosec}^2 x - 1) \, dx \quad \textcircled{1}$$

$$= -\frac{1}{3} (0-1) - (\cot x - x) \Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}}$$

$$= \frac{1}{3} - [0-1 - (\frac{\pi}{2} - \frac{\pi}{4})]$$

$$= \frac{1}{3} + 1 - \frac{\pi}{4} = \frac{\pi}{4} + \frac{4}{3} = I$$

$$\therefore \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^6 x \, dx =$$

$$\textcircled{1} \quad = \frac{13}{15} \cdot \frac{\pi}{4}$$

Q13

a) i) $k \geq 3$ $(k-1)(k+1) < k^2$

show $\frac{1}{(k-1)(k+1)} > \frac{1}{k^3}$

Since $(k-1)(k+1) < k^2$, $k \geq 3$

$\therefore (k-1)k(k+1) < k^3$

$\therefore \frac{1}{(k-1)k(k+1)} > \frac{1}{k^3}$ ①

ii) $S_n = \frac{1}{3^3} + \frac{1}{4^3} + \frac{1}{5^3} + \dots + \frac{1}{n^3} = \sum_{k=3}^n \frac{1}{k^3}$

to prove $S_n < \frac{1}{12}$

Let $\frac{1}{(k-1)k(k+1)} = \frac{A}{(k-1)k} + \frac{B}{k(k+1)} = \frac{1/2}{(k-1)k} - \frac{1/2}{k(k+1)}$ ①

Now $S_n = \frac{1}{3^3} + \frac{1}{4^3} + \frac{1}{5^3} + \dots + \frac{1}{n^3}$

$\therefore S_n < \frac{1}{2 \times 3 \times 4} + \frac{1}{3 \times 4 \times 5} + \frac{1}{4 \times 5 \times 6} + \dots + \frac{1}{(n-1)n(n+1)}$ using (i) inserted

$= \sum_{k=3}^n \left[\frac{1}{(k-1)k} - \frac{1}{k(k+1)} \right]$ frame ①

$\therefore 2S_n < \left[\frac{1}{2 \times 3} - \frac{1}{3 \times 4} + \frac{1}{3 \times 4} - \frac{1}{4 \times 5} + \frac{1}{4 \times 5} - \dots \right]$

$$\therefore \left[\cancel{\frac{1}{(n-1)n}} - \frac{1}{n(n+1)} \right] = \frac{1}{2 \times 3} - \frac{1}{n(n+1)}$$

$$\therefore 2S_n < \frac{1}{6} - \frac{1}{n(n+1)}$$

$$\therefore 2S_n < \frac{1}{6} \text{ for } n \geq 3$$

$$\therefore S_n < \frac{1}{12} \quad \textcircled{1}$$

$$Q13 b) i) \frac{1}{p} + \frac{1}{q} = 1$$

$$\therefore p+q = pq = s$$

$$(p+q)^2 = p^2 + q^2 + 2pq \quad (1)$$

$$(p+q)^2 = (pq)^2 = s^2 \quad (2)$$

①

$$(1) \& (2) \Rightarrow (p+q)^2 \geq 4pq \Rightarrow s \geq 4$$

$$\begin{aligned} \frac{1}{p(p+1)} + \frac{1}{q(q+1)} &= \frac{1}{p} - \frac{1}{p+1} + \frac{1}{q} - \frac{1}{q+1} = \\ &= 1 - \frac{p+q+2}{(p+1)(q+1)} \\ &= 1 - \frac{s+2}{2s+1} = \frac{s-1}{2s+1} \end{aligned}$$

$\therefore$ we have to prove

$$\frac{1}{3} \leq \frac{s-1}{2s+1} \leq \frac{1}{2}$$

but $2s+1 \leq 3s-3 \Leftrightarrow 4 \leq s$ and

①

$$2s-2 \leq 2s+1 \Leftrightarrow -2 \leq 1 \quad \square$$

b) ii)

$$\frac{1}{p(p-1)} + \frac{1}{q(q-1)} = \frac{1}{p-1} - \frac{1}{p} + \frac{1}{q-1} - \frac{1}{q} =$$

$$= \frac{p+q-2}{(p-1)(q-1)} - 1 \quad \textcircled{1}$$

$$= \frac{s-2}{s-1} - 1$$

$$= s-3 \text{ and as } s \geq 4 \text{ from (i)}$$

$$\therefore s-3 \geq 1$$

①

$$\therefore \frac{1}{p(p-1)} + \frac{1}{q(q-1)} \geq 1$$

Q13 c) $N = \frac{u}{s}$ show $N^2 = 0.01 \frac{x^4 - 64}{x^2}$

Diagram: A horizontal axis labeled x with a point at $x = -4$ labeled $t=0$. The origin is marked 0 . To the right, $t=0, x=4$ m is labeled $f(0)$. Below the axis, $|N| = \frac{\sqrt{3}}{5} \frac{u}{s}$ and $\therefore N = -\frac{\sqrt{3}}{5}$ are written.

$$\ddot{x} = 0.01 \left(x + \frac{64}{x^3} \right) = f(x)$$

$$\frac{d}{dx} \left(\frac{1}{2} N^2 \right) = 0.01 \left(x + \frac{64}{x^3} \right) \quad (1)$$

$$\frac{1}{2} N^2 = 0.01 \left(\frac{x^2}{2} - \frac{64}{2x^2} \right) + C$$

$$N^2 = 0.01 \left(x^2 - \frac{64}{x^2} \right) + C$$

$$x = -4, N = -\frac{\sqrt{3}}{5}$$

$$\frac{3}{25} = \frac{1}{100} \left(16 - \frac{64}{16} \right) + C$$

$$12 = 12 + C \Rightarrow C = 0$$

$$\therefore N^2 = 0.01 \left(\frac{x^4 - 64}{x^2} \right)$$

i) why $N = + \frac{\sqrt{x^4 - 64}}{10x}$

Object will be stationary when $N=0$

$$\therefore x^4 - 64 = 0$$

①

$$\therefore x = \pm 2\sqrt{2}$$

Since object starts at $x = -4$ and moves left it will never come to rest ($x = -2\sqrt{2}$),

$\therefore$ velocity is always negative, so when we take the $\sqrt{\quad}$ we need to choose an expression that is always negative and since $x < 0$

$$\therefore v = \frac{\sqrt{x^4 - 64}}{10x}$$

OR For $x < 0$, x and $\frac{1}{x^3}$ are both negative

$\therefore$ the acceleration is always negative

for $x < 0$ and since the initial direction of motion is to the left from position $x = -4$ then the object will always move towards the left, $\therefore$ the velocity will be always negative. Therefore choose the positive

$$\sqrt{x^4 - 64} \times 10x < 0$$

$$\text{iii)} \quad t = ? \quad x = -50 \text{ m}$$

$$v = \frac{\sqrt{x^4 - 64}}{10x}$$

$$\frac{dx}{dt} = \frac{\sqrt{x^4 - 64}}{10x} \quad (1)$$

$$\frac{dt}{dx} = \frac{10x}{\sqrt{x^4 - 64}}$$

$$\int_0^t dt = \int_{-4}^{-50} \frac{10x}{\sqrt{x^4 - 64}} dx$$

$$\text{let } u = x^2 \quad x = -4, \quad u = 16 \quad (1)$$

$$x = -50, \quad u = 2500$$

$$du = 2x dx$$

$$t = \int_{16}^{2500} \frac{5 du}{\sqrt{u^2 - 64}}$$

$$= 5 \left[\ln |u + \sqrt{u^2 - 64}| \right]_{16}^{2500}$$

$$= 5 \left[\ln(2500 + \sqrt{2500^2 - 64}) - \ln(16 + \sqrt{16^2 - 64}) \right]$$

RCR for

$$\int \frac{f'(x)}{\sqrt{f(x)^2 - a^2}} = \ln |f(x) + \sqrt{f(x)^2 - a^2}| + C$$

(1)

$\approx 25.604 \text{ sec}$

time, correct to nearest second, to reach a position 50m to the left of the origin is 26 seconds. ①

13(a)

$$P, (k-1)(k+1) < k^2$$

Multiplying both sides by k ($k \geq 3$ given) } or equivalent
 $(k-1)k(k+1) < k^3$ working ✓

$$\Rightarrow \frac{1}{(k-1)k(k+1)} > \frac{1}{k^3}$$

$$(ii) \text{ Let } f(k) = -\frac{1}{2k(k+1)}$$

$$\Rightarrow f(k) - f(k-1) = \frac{1}{2(k-1)k} - \frac{1}{2k(k+1)}$$

$$= \frac{(k+1) - (k-1)}{2(k-1)k(k+1)}$$

$$= \frac{1}{(k-1)k(k+1)}$$

$$\therefore \frac{1}{k^3} < \frac{1}{k(k-1)(k+1)} \Rightarrow \frac{1}{k^3} < f(k) - f(k-1) \checkmark$$

$$\Rightarrow \frac{1}{3^3} < f(3) - f(2)$$

$$\frac{1}{4^3} < f(4) - f(3)$$

$$\frac{1}{5^3} < f(5) - f(4)$$

$$\frac{1}{6^3} < f(6) - f(5) \quad \dots \quad \frac{1}{(n-2)^3} < f(n-2) - f(n-3) \quad \text{falling sum}$$

$$\frac{1}{(n-1)^3} < f(n-1) - f(n-2) \checkmark$$

$$\frac{1}{n^3} < f(n) - f(n-1)$$

$$\Rightarrow \text{Adding all } S_n < f(n) - f(2)$$

$$= -\frac{1}{2n(n+1)} - \frac{1}{2 \cdot 2 \cdot (3)}$$

$$= -\frac{1}{12} - \frac{1}{2n(n+1)} \checkmark$$

$$< -\frac{1}{12} //$$

$$13(b) \quad \frac{1}{p} + \frac{1}{q} = 1 \Rightarrow \boxed{p+q = pq} \text{ and since } \frac{1}{p} + \frac{1}{q} \geq \frac{2}{\sqrt{pq}} \text{ (AM} \geq \text{GM)} \\ \Rightarrow 1 \geq \frac{2}{\sqrt{pq}} \\ \Rightarrow \boxed{pq \geq 4}$$

$$\begin{aligned} \text{let } t &= \frac{1}{p(p+1)} + \frac{1}{q(q+1)} \\ &= \frac{1}{p} - \frac{1}{p+1} + \frac{1}{q} - \frac{1}{q+1} \\ &= 1 - \left(\frac{1}{p+1} + \frac{1}{q+1} \right) \quad (\because \frac{1}{p} + \frac{1}{q} = 1) \\ &= 1 - \frac{2+pq}{1+2pq} \quad (\because p+q = pq) \\ &= \frac{1}{2} - \frac{3}{2(1+2pq)} \end{aligned}$$

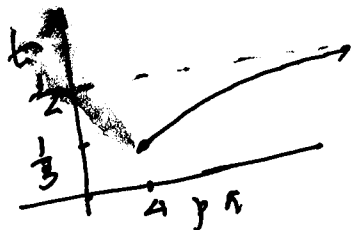
t is continuous $\forall pq$

$$\frac{dt}{d(pq)} = \frac{3}{(1+2pq)^2}$$

$\Rightarrow t$ is ~~monotonic~~ increasing w.r.t pq ✓

$$\text{when } pq=4 \quad t = \frac{1}{2} - \frac{3}{2(1+8)} = \frac{1}{2} - \frac{1}{6} = \frac{1}{3}$$

$$\text{as } pq \rightarrow \infty \quad t \rightarrow \frac{1}{2}$$



$$\Rightarrow \frac{1}{3} \leq t < \frac{1}{2}$$

$$\text{i.e., } \frac{1}{3} \leq \frac{1}{p(p+1)} + \frac{1}{q(q+1)} < \frac{1}{2}$$

$$13(b)(ii) \quad \frac{1}{p(p-1)} + \frac{1}{q(q-1)}$$

$$= \frac{1}{p-1} - \frac{1}{p} + \frac{1}{q-1} - \frac{1}{q}$$

$$= \frac{1}{p-1} + \frac{1}{q-1} - 1 \quad \checkmark \quad \left(\because \frac{1}{p} + \frac{1}{q} = 1 \right)$$

$$= \frac{pq-2}{pq-p-q+1} - 1$$

$$= \frac{pq-2}{1} - 1 \quad \checkmark$$

$$= pq-3$$

$$\geq 1 \quad (\text{since } pq \geq 4 \text{ from (i)})$$

$$13(c) \quad (i) \quad \ddot{x} = 0.01 \left(x + \frac{64}{x^3} \right)$$

$$\frac{d(\frac{1}{2}v^2)}{dx} = 0.01 \left(x + \frac{64}{x^3} \right) \quad \checkmark$$

$$\int d(\frac{1}{2}v^2) = \int 0.01 \left(x + \frac{64}{x^3} \right) dx$$

$$\frac{v^2}{2} = 0.01 \left(\frac{x^2}{2} - \frac{32}{x^2} \right) + C$$

$$\text{Initially } x = -4, v = -\frac{\sqrt{3}}{5}$$

$$\Rightarrow C = \frac{3}{50} - 0.01(8-2) = 0 \quad \checkmark$$

$$\Rightarrow v^2 = \frac{0.01}{x^2} (x^4 - 64)$$

$$v = \pm \frac{\sqrt{x^4 - 64}}{10x}$$

$$\text{Initially } v < 0, x < 0 \Rightarrow v = -\frac{\sqrt{x^4 - 64}}{10x} \quad \checkmark$$

$v = -\frac{\sqrt{x^4 - 64}}{10x}$ is not acceptable as it contradicts the initial condition

13(c) (iii)

$$V = \frac{\sqrt{x^2 - 64}}{10x}$$

$$\frac{dx}{dt} = \frac{\sqrt{x^2 - 64}}{10x}$$

$$\Rightarrow \int_{-4}^{50} \frac{10x}{\sqrt{x^2 - 64}} dx = \int_0^t dt$$

$$\Rightarrow \left[5 \ln \left[x^2 + \sqrt{x^2 - 64} \right] \right]_{-4}^{50} = t$$

$$\Rightarrow t = 26.220 \dots$$

≈ 26 min

Q14 a) $a_1, a_2, a_3, \dots, a_n \in \mathbb{R}^+$ ✓

$$a_1 \times a_2 \times a_3 \times \dots \times a_n = 1$$

Prove $(a_1^2 + a_1)(a_2^2 + a_2) \dots (a_n^2 + a_n) \geq 2^n$

$$a_1 \geq \sqrt{a_1}$$

$$(a_1 - \sqrt{a_1})^2 \geq 0$$

$$a_1^2 - 2a_1\sqrt{a_1} + a_1 \geq 0$$

$$a_1^2 + a_1 \geq 2a_1\sqrt{a_1} \quad (1)$$

Similarly for a_2 to a_n

$$a_2^2 + a_2 \geq 2a_2\sqrt{a_2} \quad (2)$$

①

$$\underline{a_n^2 + a_n \geq 2a_n\sqrt{a_n} \quad (n)} \quad \text{①}$$

$$(1) \times (2) \times \dots \times (n) :$$

$$\therefore (a_1^2 + a_1)(a_2^2 + a_2)(a_3^2 + a_3) \dots (a_n^2 + a_n) \geq 2 \times a_1\sqrt{a_1} \times 2a_2\sqrt{a_2} \times \dots \times 2a_n\sqrt{a_n}$$

$$\geq 2^n (a_1 a_2 \dots a_n) \sqrt{a_1 a_2 \dots a_n}$$

①

$$\geq 2^n$$

Q14 b)

i) $m = 90 \text{ kg}$ $h = 100 \text{ m}$ $g = 10 \text{ m/s}^2$
 $v = f(x) = ?$ $u = 0$

d) no resistance

$\begin{array}{c} \downarrow t=0 \\ \downarrow t \\ \downarrow x \\ \downarrow v \end{array}$ $\downarrow mg$

$$m\ddot{x} = mg$$
$$\ddot{x} = g$$
$$v \frac{dv}{dx} = g$$

$$\int v dv = \int 10 dx$$
$$\frac{v^2}{2} \Big|_u^v = 10x \Big|_0^x$$

$$\frac{v^2 - u^2}{2} = 10x$$

$$\frac{v^2}{2} = 10x + \frac{u^2}{2}$$

$$v^2 = 20x + u^2 \Rightarrow v = \sqrt{u^2 + 20x}, \quad v > 0$$

$$b) \ddot{m}x = mg - 0.27 v^2$$

$$\ddot{x} = g - \frac{0.27}{90} v^2$$

$$\ddot{x} = 10 - 0.003 v^2 \quad (1) \quad (*)$$

$$v \frac{dv}{x dx} = 10 - 0.003 v^2$$

$$\int_0^x dx = \int_0^v \frac{v dv}{10 - 0.003 v^2}$$

$$x = -\frac{1}{0.003} \frac{1}{2} \left[\ln |10 - 0.003 v^2| \right]_0^v \quad (1)$$

$$= -\frac{1}{0.006} \ln \frac{10 - 0.003 v^2}{10 - 0.003 u^2}$$

$$\ln \frac{10 - 0.003 v^2}{10 - 0.003 u^2} = -0.006 x$$

$$\frac{10 - 0.003 v^2}{10 - 0.003 u^2} = e^{-0.006 x}$$

$$(10 - 0.003 v^2) = (10 - 0.003 u^2) e^{-0.006 x} \quad (1)$$

$$v = \sqrt{\frac{10 - (10 - 0.003 u^2) e^{-0.006 x}}{0.003}}, \quad v \geq 0$$

5) ii) $t=0$, $v=0$, 60 m from the ground

(2) $v = \dot{x} = ?$
after 40m

$$x = 40 \text{ m}$$

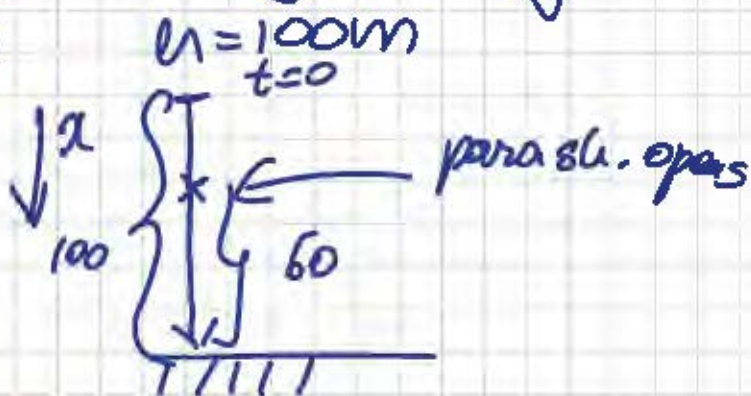
from i) (2):

$$v = \sqrt{u^2 + 20x}$$

$$v = \sqrt{0^2 + 20 \times 40}$$

$$= 20\sqrt{2} \text{ m/s}$$

$$\approx 28.28 \text{ m/s}$$



10) $v = \dot{x} = ?$ on landing, $x=60 \text{ m}$, $u=20\sqrt{2}$
from i) (b) (now with resistance)

$$v = \sqrt{\frac{10 - (10 - 0.003 \times (20\sqrt{2})^2) e^{-0.006 \times 60}}{0.003}}$$

$$v = \sqrt{\frac{10 - (10 - 0.003 \times 800) e^{-0.086}}{0.003}}$$

$$v = \sqrt{\frac{10 - 5.3023}{0.003}} = \sqrt{1565.8866}$$

$$v = 39.571285 \approx 39.6 \text{ m/s}$$

$$b) \quad v_L = 39.6 \quad (\text{from } b)(v))$$

$$\frac{v_L}{v_T} = ?$$

$$v_T = ? \quad \text{when } \ddot{x} = 0$$

$$\text{from } (*) (i) (v) \quad \ddot{x} = 10 - 0.003 v^2 \Rightarrow$$

$$v_T = \sqrt{\frac{10}{0.003}}$$

①

$$v_T = 57.73502 \quad \frac{\text{m}}{\text{s}}$$

$$= 57.7 \quad \frac{\text{m}}{\text{s}}$$

$$\frac{v_L}{v_T} = \frac{39.6}{57.7}$$

$$100\% = 68.539\%$$

$$\approx 69\% \quad \text{①}$$

14 c)

$$a, b, c \in \mathbb{Z}^+$$

$a^2 + b + c$, $b^2 + a + c$, $c^2 + a + b$ can't be perfect squares

let assume $a \leq b \leq c$

$$\therefore c^2 < c^2 + a + b \leq c^2 + 2c < (c+1)^2$$

$$\therefore c^2 + a + b < (c+1)^2$$

$\therefore c^2 + a + b$ cannot be perfect square

Similarly for the other two.

Q15

a) $r = (2+\lambda)\hat{i} + (3-\lambda)\hat{j} + (4-2\lambda)\hat{k}$

i) in xy plane: $z=0$, $4-2\lambda=0 \Rightarrow \lambda=2$

point A (4, 1, 0) ①

in xz plane: $y=0$, $3-\lambda=0 \Rightarrow \lambda=3$

point B (5, 0, -2) ①

in yz plane: $x=0$, $2+\lambda=0 \Rightarrow \lambda=-2$

point C (0, 5, 8) ①

iii) $\text{proj}_{\vec{AB}} \vec{OC} = \frac{\vec{OC} \cdot \vec{AB}}{|\vec{AB}|^2} \vec{AB}$

$\vec{AB} = \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix}$

$\vec{OC} = \begin{pmatrix} 0 \\ 5 \\ 8 \end{pmatrix}$

$= \frac{(0 - 5 - 16)}{(\sqrt{1+1+4})^2} \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix}$

$= \frac{-21}{6} \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix}$

$= -\frac{7}{2} \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix} = \vec{XC}$

iv)

if not used collinearity of A, B, C

$\vec{OX} + \vec{XC} = \vec{OC}$

$\therefore \vec{OX} = \vec{OC} - \vec{XC}$

$= \begin{pmatrix} 0 \\ 5 \\ 8 \end{pmatrix} + \frac{7}{2} \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix}$

Area $\Delta OAC = \frac{1}{2} |\vec{AC}| \times |\vec{OX}|$

$\vec{AC} = \begin{pmatrix} 7 \\ 4 \\ 4 \end{pmatrix}$

$\vec{OX} = \begin{pmatrix} 7/2 \\ 5 - 7/2 \\ 8 - 7 \end{pmatrix} = \begin{pmatrix} 7/2 \\ 3/2 \\ 1 \end{pmatrix}$

$= \frac{1}{2} \begin{pmatrix} 7 \\ 3 \\ 2 \end{pmatrix}$

$$\begin{aligned}
 \text{Area } \triangle OAC &= \frac{1}{2} \sqrt{16+16+64} \times \sqrt{\frac{1}{4}(49+9+9)} \\
 &= \frac{1}{2} \sqrt{98} \sqrt{\frac{1}{4} 62} \\
 &= \frac{1}{4} 4\sqrt{6} \sqrt{62} \\
 &= \sqrt{3 \times 2 \times 2 \times 31}
 \end{aligned}$$

point to use part
iii)

$$= 2\sqrt{93}$$

$$= 19.28730152$$

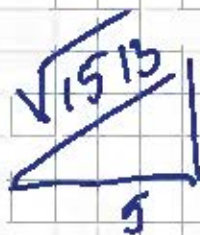
$$\approx 19.3 \text{ (3 s.f.)} \quad \textcircled{1}$$

$$\text{or } A = \frac{1}{2} |\vec{OA}| |\vec{OC}| \sin(\angle COA)$$

$$\begin{aligned}
 \vec{OC} \cdot \vec{OA} &= 5 = |\vec{OA}| |\vec{OC}| \cos(\angle COA) \\
 &= \sqrt{17} \sqrt{89} \cos(\angle COA)
 \end{aligned}$$

$$\cos(\angle COA) = \frac{5}{\sqrt{1513}} \Rightarrow \sin(\angle COA) = \frac{\sqrt{1488}}{\sqrt{1513}}$$

$$\begin{aligned}
 \therefore A &= \frac{1}{2} \sqrt{1513} \frac{\sqrt{1488}}{\sqrt{1513}} = 2\sqrt{93} \approx 19.3
 \end{aligned}$$



$$15b) I_n = \int_0^{\frac{\pi}{2}} (\sin x)^n dx, n \geq 0$$

$$1) n \geq 2, I_n = \left(\frac{n-1}{n}\right) I_{n-2}$$

$$I_n = \int_0^{\frac{\pi}{2}} (\sin x)^{n-1} \frac{d}{dx} (-\cos x) dx$$

use only trig.
fun. when int-
by parts

$$= \left[(\sin x)^{n-1} \cdot (-\cos x) \right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} (-\cos x) \frac{d}{dx} (\sin x)^{n-1} dx$$

$$= [0 - 0] + (n-1) \int_0^{\frac{\pi}{2}} (\sin x)^{n-2} \cos x \cos x dx$$

$$= (n-1) \int_0^{\frac{\pi}{2}} (\sin x)^{n-2} \cos^2 x dx$$

$$= (n-1) \int_0^{\frac{\pi}{2}} (\sin x)^{n-2} (1 - \sin^2 x) dx$$

$$= (n-1) \int_0^{\frac{\pi}{2}} (\sin x)^{n-2} dx - (n-1) \int_0^{\frac{\pi}{2}} \sin^n x dx$$

$$\therefore I_n = (n-1) I_{n-2} - (n-1) I_n$$

$$\therefore I_n (1 + (n-1)) = (n-1) I_{n-2}$$

$$\therefore I_n = \frac{n-1}{n} I_{n-2}, n \geq 2 \quad \square$$

$$\begin{aligned}
 \text{ii) from i) } I_{2n} &= \left(\frac{2n-1}{2n} \right) I_{2n-2} \\
 &= \frac{2n-1}{2n} \left(\frac{2n-3}{2n-2} I_{2n-4} \right) \text{ re-using the } i) \text{ results} \\
 &= \frac{2n-1}{2n} \frac{2n-3}{2n-2} \frac{2n-5}{2n-4} I_{2n-6} \\
 &= \frac{2n-1}{2n} \frac{2n-3}{2n-2} \frac{2n-5}{2n-4} \dots \frac{(2n-2n)+1}{(2n-2n)+1} I_{2n-2n} \\
 &= \frac{(2n-1)(2n-3)(2n-5) \dots 1}{(2n)(2n-2)(2n-4) \dots 2} I_0
 \end{aligned}$$

①

$$I_0 = \int_0^{\pi/2} (\sin x)^0 dx = x \Big|_0^{\pi/2} = \frac{\pi}{2}$$

$$\therefore I_{2n} = \frac{2n-1}{2n} \cdot \frac{2n-3}{2n-2} \cdot \frac{2n-5}{2n-4} \dots \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2}$$

①

$$\text{Also, from (i) } I_{2n+1} = \frac{2n}{2n+1} I_{2n-1}$$

$$\begin{aligned}
 &= \frac{2n}{2n+1} \left(\frac{2n-2}{2n-1} I_{2n-3} \right) \text{ using (i) again} \\
 &= \frac{2n}{2n+1} \frac{2n-2}{2n-1} \frac{2n-4}{2n-3} I_{2n-5} \text{ again (i)}
 \end{aligned}$$

$$= \frac{2n}{2n+1} \cdot \frac{2n-2}{2n-1} \cdot \frac{2n-4}{2n-3} \cdots \frac{2}{3} \cdot \frac{1}{1}$$

$$I_1 = \int_0^{\pi/2} \sin x \, dx = -\cos x \Big|_0^{\pi/2}$$

$$= -(0-1) = 1$$

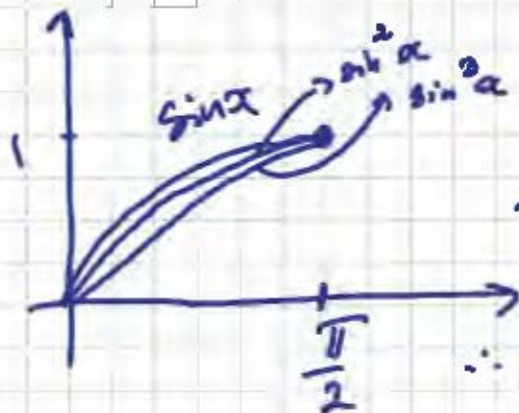
①

$$\therefore I_{2n+1} = \frac{2n}{2n+1} \cdot \frac{2n-2}{2n-1} \cdot \frac{2n-4}{2n-3} \cdots \frac{4}{5} \cdot \frac{2}{3} \cdot 1$$

$$\text{iii) } I_k = \int_0^{\pi/2} (\sin x)^k \, dx \quad k \geq 0$$

$$I_{k+1} = \int_0^{\pi/2} (\sin x)^{k+1} \, dx$$

I_k & I_{k+1} represent the area under the curves $y = (\sin x)^k$ & $y = (\sin x)^{k+1}$ for $0 \leq x \leq \frac{\pi}{2}$



for $0 \leq x \leq \frac{\pi}{2}$

$$0 \leq \sin x \leq 1$$

$$\therefore (\sin x)^{k+1} \leq (\sin x)^k$$

$\therefore$ area under the

Curve y_k is greater than the area under the curve y_{k+1} for $0 \leq x \leq \frac{\pi}{2}$
 $\therefore I_k > I_{k+1}$ ①

iv) using the results in (ii) for I_{2n+1}

$$I_{2n-1} = \frac{2n-2}{2n-1} \cdot \frac{2n-4}{2n-3} \cdot \frac{2n-6}{2n-5} \cdots \frac{4}{5} \cdot \frac{2}{3} \cdot 1$$

from (ii), $I_k > I_{k+1}$ and $I_{2n-1} > I_{2n}$ and $I_{2n-1} > I_{2n+1}$

Since $I_{2n-1} > I_{2n}$, then

$$\frac{2n-2}{2n-1} \cdot \frac{2n-4}{2n-3} \cdots \frac{4}{5} \cdot \frac{2}{3} \cdot 1 > \frac{2n-1}{2n} \cdot \frac{2n-3}{2n-2} \cdots \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2}$$

Thus:

$$\frac{(2n)(2n-2)^2(2n-4)^2 \cdots 4^2 \cdot 2^2}{(2n-1)^2(2n-3)^2 \cdots 5^2 \cdot 3^2 \cdot 1} > \frac{\pi}{2} \quad \text{①}$$

hence

$$\frac{2n}{2n+1} \cdot \frac{2n(2n-2)^2(2n-4)^2 \cdots 4^2 \cdot 2^2}{(2n-1)^2(2n-3)^2 \cdots 5^2 \cdot 3^2 \cdot 1} > \frac{2n}{2n+1} \cdot \frac{\pi}{2}$$

i.e.
$$\frac{2^2 \cdot 4^2 \cdots (2n)^2}{1 \cdot 3^2 \cdot 5^2 \cdots (2n-1)^2 (2n+1)} > \frac{2n}{2n+1} \cdot \frac{\pi}{2} \quad (1)$$

Since $I_{2n} > I_{2n+1}$, then

$$\underbrace{\frac{2n-1}{2n} \cdot \frac{2n-3}{2n-2} \cdot \frac{2n-5}{2n-4} \cdots \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2}}_{(*)} > \frac{2n}{2n+1} \cdot \frac{2n-2}{2n-1} \cdot$$

$$(*) \frac{\pi}{2} > \frac{\frac{2n-4}{2n-3} \cdots \frac{4}{5} \cdot \frac{2}{3} \cdot 1 \cdot (2n)^2 (2n-2)^2 \cdots 4^2 \cdot 2^2}{(2n+1) (2n-1)^2 (2n-3)^2 \cdots 3^2 \cdot 1}$$

$$(*) \frac{\pi}{2} > \frac{2^2 \cdot 4^2 \cdots (2n)^2}{1 \cdot 3^2 \cdot 5^2 \cdots (2n-1)^2 (2n+1)} \quad (2)$$

Now from (1) and (2)

$$\therefore \frac{\pi}{2} \left(\frac{2n}{2n+1} \right) < \frac{2^2 \cdot 4^2 \cdots (2n)^2}{1 \cdot 3^2 \cdot 5^2 \cdots (2n-1)^2 (2n+1)} < \frac{\pi}{2}$$

or to look at

$$I_{2n+1} \cdot I_{2n} < I_{2n+1} I_{2n-1} < I_{2n} I_{2n-1}$$

Q16 a)

$$z = \frac{1}{4} (1 + \sqrt{3}i)$$

$$i) |1 + \sqrt{3}i| = \sqrt{1+3} = \sqrt{4} = 2 \quad (1)$$

$$\arg(1 + \sqrt{3}i) = \tan^{-1} \frac{\sqrt{3}}{1} = \frac{\pi}{3}$$

$$\therefore z = \frac{1}{4} (2 e^{i\pi/3}) = \frac{1}{2} e^{i\pi/3} \quad (1)$$

$$ii) z^2 = \left(\frac{1}{2} e^{i\pi/3}\right)^2 = \frac{1}{4} e^{i2\pi/3} \quad (1)$$

$$iii) z^3 = \left(\frac{1}{2} e^{i\pi/3}\right)^3 = \frac{1}{8} e^{i\pi} \quad (1)$$

$$Ar \Delta OP_0P_1 = \frac{1}{2} \times (1) \times \left(\frac{1}{2}\right) \sin \frac{\pi}{3} \quad (1)$$

$$= \frac{\sqrt{3}}{8} \text{ units}^2$$

$$Ar OP_1P_2 = \frac{1}{2} \times \left(\frac{1}{2}\right) \times \left(\frac{1}{4}\right) \sin \frac{\pi}{3} = \frac{\sqrt{3}}{32} \text{ units}^2 \quad (1)$$

$$Ar \Delta OP_2P_3 = \frac{1}{2} \times \left(\frac{1}{4}\right) \times \left(\frac{1}{8}\right) \times \sin \frac{\pi}{3} = \frac{\sqrt{3}}{128} \text{ units}^2 \quad (1)$$

$$\therefore \text{Area of } (3+1)\text{-polygon} = \frac{\sqrt{3}}{8} + \frac{\sqrt{3}}{32} + \frac{\sqrt{3}}{128} = \frac{21\sqrt{3}}{128} \text{ units}^2 \quad (1)$$

$$\begin{aligned}
 & \text{iv) } \frac{1}{2} (1) \left(\frac{1}{2}\right) \sin \frac{\pi}{n} + \frac{1}{2} \left(\frac{1}{2}\right) \left(\frac{1}{4}\right) \sin \frac{\pi}{n} + \\
 & + \frac{1}{2} \left(\frac{1}{4}\right) \left(\frac{1}{8}\right) \sin \frac{\pi}{n} \dots + \textcircled{1} \frac{1}{2} \left(\frac{1}{2^{n-1}}\right) \left(\frac{1}{2^n}\right) \sin \frac{\pi}{n} \\
 & = \frac{1}{2} \left(1 \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{4} + \frac{1}{4} \times \frac{1}{8} + \dots + \frac{1}{2^{n-1}} \times \frac{1}{2^n} \right) \sin \frac{\pi}{n} \\
 & = \textcircled{1} \frac{1}{2} \left(\frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \dots + \frac{1}{2^{2n-1}} \right) \sin \frac{\pi}{n}
 \end{aligned}$$

n terms

Area of $(n+1)$ -polygon :

$$\begin{aligned}
 S_n &= a \frac{1-r^n}{1-r} \\
 \therefore &= \frac{1}{2} \left(\frac{\frac{1}{2} \left(1 - \left(\frac{1}{2^2}\right)^n \right)}{1 - \frac{1}{2^2}} \right) \sin \frac{\pi}{n}
 \end{aligned}$$

$$= \frac{1}{4} \frac{4}{3} \left(1 - \left(\frac{1}{4}\right)^n \right) \sin \frac{\pi}{n}$$

$$= \frac{1}{3} \left(1 - \left(\frac{1}{4}\right)^n \right) \sin \frac{\pi}{n}$$

$$\therefore a = \frac{1}{3}, b = \frac{1}{4}$$

\textcircled{1}

(C6) i)

$\vec{A} = \frac{b}{|b|} + \frac{c}{|c|}$ ①

ii)

Set origin at A:

- $\vec{AB} = \vec{OB} = b$
- $\vec{AC} = \vec{OC} = c$
- $\vec{AD} = \vec{OD} = d$
- $\vec{AE} = \vec{OE} = e$
- $\vec{AG} = \vec{OG} = g$
- $\vec{AF} = \vec{OF} = f$

$DE \parallel CB$

$DF : FC = EG : GD = BE : CD$

Show $GF \parallel AX$

$\vec{c} = p \vec{b}$ (1)
 $\vec{d} = q \vec{c}$ (2)
 $\left. \begin{matrix} (1) \\ (2) \end{matrix} \right\} p, q \in (0,1)$
 let $\frac{BF}{FC} = t$

$$\left. \begin{aligned} \vec{BF} + \vec{b} &= \vec{f} \Rightarrow t\vec{FC} + \vec{b} = \vec{f} \\ \vec{FC} + \vec{f} &= \vec{c} \end{aligned} \right\} \Rightarrow t(\vec{c} - \vec{f}) + \vec{b} = \vec{f}$$

$$\therefore t\vec{c} - t\vec{f} + \vec{b} = \vec{f} \Rightarrow \vec{f} = \frac{t\vec{c} + \vec{b}}{1+t} \quad (3) \quad \textcircled{1}$$

$$\text{also } \frac{\vec{EG}}{\vec{GD}} = t \Rightarrow \vec{EG} = t\vec{GD}$$

$$\left. \begin{aligned} \vec{e} + \vec{EG} &= \vec{g} \Rightarrow \vec{e} + t\vec{GD} = \vec{g} \\ \vec{g} + \vec{GD} &= \vec{d} \end{aligned} \right\} \Rightarrow \vec{e} + t(\vec{d} - \vec{g}) = \vec{g}$$

$$\therefore \vec{e} + t\vec{d} - t\vec{g} = \vec{g} \Rightarrow \vec{g} = \frac{\vec{e} + t\vec{d}}{1+t} \text{ then from}$$

$$(1) \& (2) \quad \vec{g} = \frac{p\vec{b} + t\vec{g}\vec{c}}{1+t} \quad (4) \quad \textcircled{1}$$

$$\text{also } \frac{|\vec{BE}|}{|\vec{CD}|} = t \Rightarrow |\vec{BE}| = t|\vec{CD}|$$

$$|\vec{BE}| = |\vec{b} - \vec{e}| = |\vec{b} - p\vec{b}| = (1-p)|\vec{b}|$$

$$|\vec{c}| = |\underline{c} - q| = |\underline{c} - q\underline{c}| = (1-q)|\underline{c}|$$

$$\therefore (1-p)|\underline{b}| = t(1-q)|\underline{c}| \quad (5)$$

then using (3) and (4)

$$\vec{GF} = \underline{f} - q = \frac{t\underline{c} + b}{1+t} - \frac{p\underline{b} + tq\underline{c}}{1+t}$$

$$\underline{f} - q = \frac{t - tq}{1+t} \underline{c} + \frac{1-p}{1+t} \underline{b}$$

$$= \frac{t(1-q)}{1+t} \underline{c} + \frac{1-p}{1+t} \underline{b}$$

$$\text{and (5)} \Rightarrow t(1-q) = \frac{(1-p)|\underline{b}|}{|\underline{c}|}$$

$$\therefore \underline{f} - q = \frac{(1-p)|\underline{b}|}{1+t} \frac{1}{|\underline{c}|} \underline{c} + \frac{1-p}{1+t} \underline{b}$$

$$\vec{GF} = \frac{1-p}{1+t} |\underline{b}| \left(\frac{\underline{c}}{|\underline{c}|} + \frac{\underline{b}}{|\underline{b}|} \right) \quad \textcircled{1}$$

$$\therefore \vec{GF} = \text{const.} \cdot \vec{AX} \therefore \vec{GF} \parallel \vec{AX} \quad \square$$